Predicting Hotel Booking Cancellations

Improving Revenue Management and Customer Satisfaction

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PROBLEM STATEMENT & MOTIVATION

Hotel booking cancellations present a significant challenge to the hospitality industry, affecting revenue management, staffing, and inventory control. Cancellations can lead to lost revenue and underutilized resources while overbooking to compensate can result in customer dissatisfaction. Accurately predicting cancellations enables hotels to maximize efficiency by implementing proactive strategies, such as dynamic pricing, personalized promotions, and optimal overbooking policies.

The primary objective of this project is to develop a predictive model that accurately forecasts hotel booking cancellations by analyzing a comprehensive dataset of hotel bookings. We aim to identify key factors influencing cancellations, such as lead time, customer demographics, booking channels, and previous cancellation history. By leveraging data mining and machine learning techniques, we aim to provide actionable insights to help hotels minimize cancellations and optimize operations.

Understanding and predicting cancellation behavior contributes to improved revenue management and enhances customer satisfaction by ensuring that rooms are available when and where they are needed. This project aspires to bridge the gap between data analysis and practical application in the hospitality industry.

1 Literature Survey

There are a limited number of studies published on predicting booking cancellations in the hospitality industry using hotel specific data. One earlier publication is “Forecasting cancellation rates for services booking revenue management using data mining” (2010) by D. Morales and J. Wang [1]. This study attempts to forecast cancellations using two-class probability estimation methods. Most notably, they found that important drivers of cancellations are time-dependent within what they call the booking horizon. This means that an attribute which may be a high predictor of cancellation for a booking with a longer lead time may not be a predictor for a booking with a short lead time. Instead, the high predictor variable for the booking with the short lead time is a different attribute.

In addition, “Modeling the cancellation behavior of hotel guests” (2018) by M. Falk and M. Vieru [2] elucidates on factors that contribute to booking cancellations. This analysis looked at individual-level booking data and found a higher probability of cancellation based on booking method, lead time, and country of origin. This work was limited by only analyzing data from a single hotel chain in a specific country and did not include additional attributes which may be of interest to affecting cancellations such as individual characteristics and previous stay history. This study provides some interesting insights into the attributes that may be influential in predicting cancellations to include in our analysis.

The bulk of research on this specific topic is from three co-authors: Nuno Antonio, Ana Almeida, and Luis Nunes. In one of their earlier publications “Using data science to predict hotel booking cancellations,” (2016) [3] they showed it is possible to build models for predicting hotel booking cancellations with high accuracy in results. Building on this work, they later published "Predicting hotel bookings cancellation with a machine learning classification model" (2017) [4].

This study employed sophisticated machine learning classification models, such as Random Forests and Gradient Boosting Machines, to predict hotel booking cancellations and analyzed the factors influencing these cancellations. The authors found that variables like lead time, market segment, and previous cancellations significantly impact the likelihood of a booking being canceled. By implementing a machine learning model, the authors showed that predicting booking cancellations is possible from an empirical standpoint.

While we aim to learn from their research methodologies and findings, our project seeks to extend this work by exploring model interpretability and the practical implications of the predictive factors. Additionally, we intend to investigate whether incorporating new features or alternative modeling techniques can improve prediction accuracy and provide deeper insights into customer behavior.

Other studies have also explored cancellation prediction using different datasets and methodologies. For instance, some have focused on the impact of economic indicators on hotel cancellations, while others have examined the role of customer reviews and sentiment analysis. Our project differentiates itself by concentrating on the dataset provided by the two Portuguese hotels and by emphasizing actionable insights for hotel management.

2 Proposed Work

Our approach distinguishes itself by emphasizing both predictive performance and model interpretability. While previous studies focused on achieving high accuracy with complex models, we aim to balance accuracy with insights that are easily understandable by hotel management. By using interpretability tools like SHAP values, we intend to provide clear explanations of how each feature influences cancellation risk, facilitating better decision-making and strategy development. Our project involves several key stages: data preprocessing, exploratory data analysis (EDA), model development, evaluation, comparison with prior work and deployment considerations. We have explained the work needed for each of these stages below.

2.1 Data Preprocessing

Data preprocessing will be primarily in data cleaning and feature engineering. For data cleaning, missing values in critical columns will need to be addressed. For example, in critical columns like is_canceled and lead_time, missing values will be handled by removing incomplete records or imputing missing values using statistical methods. Further, outliers in numerical features such as lead_time and adr (average daily rate) must be detected and handled to prevent skewed results. Feature engineering will require encoding categorical variables using one-hot encoding or label encoding to make them suitable for machine learning algorithms. In addition, date-related features may need to be converted into numerical formats or to extract meaningful components like day of the week or season. Lastly, we will create new features if potential patterns are identified during exploratory analysis.

2.2 Exploratory Data Analysis (EDA)

To understand data distributions, identify correlations, and uncover patterns related to cancellations, exploratory data analysis (EDA) is a crucial part of our process. We will employ various visualization tools such as histograms, box plots, scatter plots, and heatmaps to examine relationships between variables and the target outcome. We will analyze the distributions of key numerical variables like lead time, average daily rate (ADR), and number of previous cancellations to detect any skewness or outliers that may affect the modeling process. For categorical variables such as hotel type, market segment, and distribution channel, we will use bar charts and count plots to assess their frequency distributions and relationships with the cancellation status. Particular attention will be given to ensuring the predictor variables chosen are not multicollinear. This will be done by calculating correlation matrices for numerical variables to identify highly correlated pairs. We will compute the Pearson correlation coefficient for numerical variables and use heatmaps to visualize these correlations. For categorical variables, we may use Cramér's V to assess associations. Additionally, we will calculate the Variance Inflation Factor (VIF) for each predictor variable to quantitatively assess multicollinearity. Variables with a high VIF value (typically above 5 or 10) indicate significant multicollinearity and may be considered for removal or transformation. We aim to improve the model's interpretability and predictive performance by addressing multicollinearity.

Through EDA, we intend to select the most relevant features, understand the underlying data patterns, and ensure that the data meets the assumptions required for our chosen modeling techniques.

2.3 Model Development

Model development will consist of three aspects: algorithm selection, hyperparameter tuning, and model interpretation. For algorithm selection, we will experiment with various classification algorithms in order to determine the best one for our model. These classification algorithms include: logistic regression, decision trees, random forests, gradient boosting machines (e.g. XGBoost) and support vector machines. In order to optimize model performance we will implement hyperparameter tuning using techniques like Grid Search or Random Search combined with cross-validation. To understand the contribution of each feature to the model's predictions for our model interpretation, we will employ methods such as feature importance analysis and SHAP (SHapley Additive exPlanations) values.

2.4 Evaluation

Next, we will evaluate the models using metrics like accuracy, precision, recall, F1-score, and ROC-AUC on a hold-out test set. We can also evaluate some models by applying k-fold cross-validation to assess generalizability. The final piece of evaluation will be to compare these performance metrics of different models to select the most effective one.

Another piece of evaluation will be a comparison with prior work. This will involve a discussion of how our findings align or differ from previous studies. This will be an opportunity to highlight any improvements in prediction accuracy or interpretability we find.

2.5 Deployment Considerations

The last phase of work will be to explore the potential integration of the predictive model into hotel reservation systems for real-time cancellation prediction.

3 Dataset Details

We are using the hotel booking demand dataset found at: https://www.kaggle.com/datasets/mojtaba142/hotel-booking/data

This data was provided by two hotels in Portugal: a resort hotel in the Algarve region (H1) and a city hotel in Lisbon (H2). The authors of the paper, Nuno Antonio, Ana de Almeida, and Luis Nunes collected and published the data with the cooperation of the hotels. The data includes 119,390 objects and 36 attributes. These attributes are a variety of data types including nominal, ordinal, binary, interval-scaled, and ratio-scaled. For instance, the field is_canceled is a binary data type field where 1 means the reservation was canceled. We also anticipate the field lead_time to be an important attribute which is the number of days between booking to reservation date, a ratio-scaled integer field. The dataset has been downloaded on both Elaine’s and Seiji’s local machine.

4 Evaluation Methods

Our evaluation methods are three main components: evaluating performance metrics, implementing validation techniques, and comparison to baseline models.

The performance metrics we foresee needing include accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix. The accuracy measures the overall correctness of the model and precision indicates the proportion of true positive predictions among all positive predictions. The recall (sensitivity) reflects the model's ability to identify actual cancellations. In order to provide balance between the precision and recall we will also use F1-Score. The ROC-AUC evaluates the model's ability to distinguish between classes across all thresholds. Finally, the confusion matrix will offer insight into true positives, false positives, true negatives, and false negatives, to further assess the accuracy of the model. In addition, if it is found that class imbalances exist in the data, this will be addressed by threshold moving or oversampling the minority such as SMOTE to ensure the model is effective and fair.

Validation techniques include K-fold cross validation and hold-out test set. The K-fold cross validation is used to ensure model stability and generalizability. For final model evaluation to assess performance on unseen data, we anticipate using the hold-out test set technique.

Lastly, is using baselines models as a method for evaluation. This method compares advanced models against simple baseline models like predicting the majority class to demonstrate improvement.

5 Tools

The project will use the Python programing language for all aspects of data analysis and modeling. The key libraries needed will be Pandas for data manipulation and preprocessing, NumPy for numerical computations, Scikit-learn for machine learning algorithms and evaluation metrics and Matplotlib and Seaborn for data visualization during EDA. Jupyter Notebook will be used as the development environment for its ability to provide interactive coding and documentation. Lastly, the tool for version control to be utilized is GitHub for collaboration and effective code management.

6 Milestones

Our project milestones align with the course deliverables and their respective due dates, outlined below.

6.1 Completed Milestones

1. By October 28:
    a. Finalize data cleaning, preprocessing and began feature engineering.
    b. Conduct initial exploratory data analysis (EDA) to understand data distributions and key patterns.
    c. Document our progress, challenges faced, and initial findings in the Progress Report (Part 3).

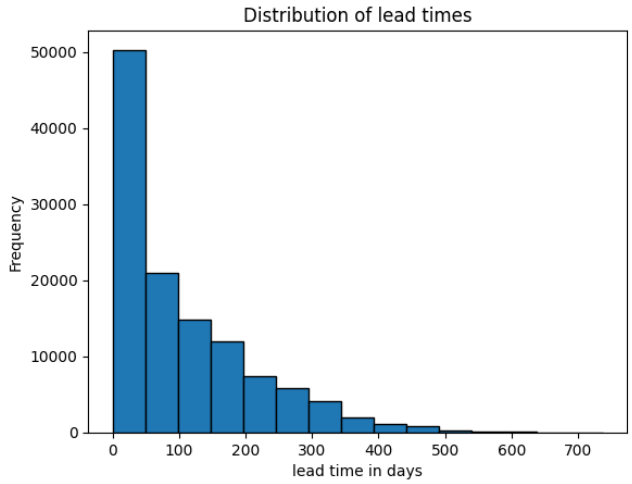
6.2 Remaining Milestones

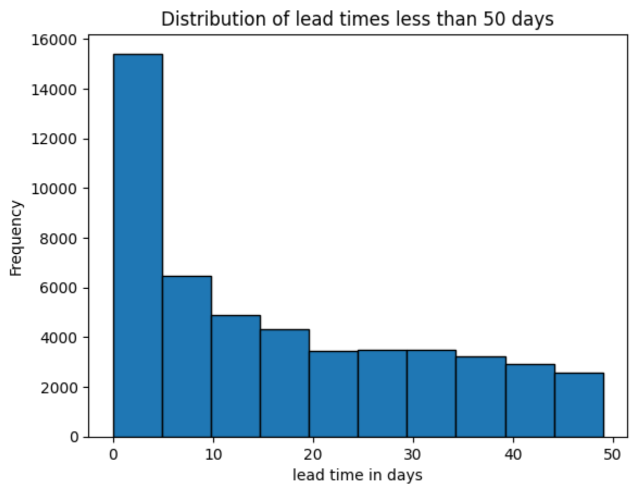
2. By November 11:
    a. Conclude EDA.
    b. Begin Preliminary modeling efforts.
3. By November 25:
    a. Conclude modeling creation.
    b. Evaluate and select the best-performing predictive model.
    c. Begin to interpret model results and extract actionable insights for hotel management.
4. By December 9:
    a. Compile all findings, methodologies, and conclusions into the Final Report (Part 4).
    b. Organize and document the project code with clear descriptions (Part 5).
    c. Prepare and practice the project presentation (Part 6).
    d. Complete peer evaluations and any required interview questions (Part 7).
    e. Submit all deliverables as per the project guidelines.

7 Results So Far

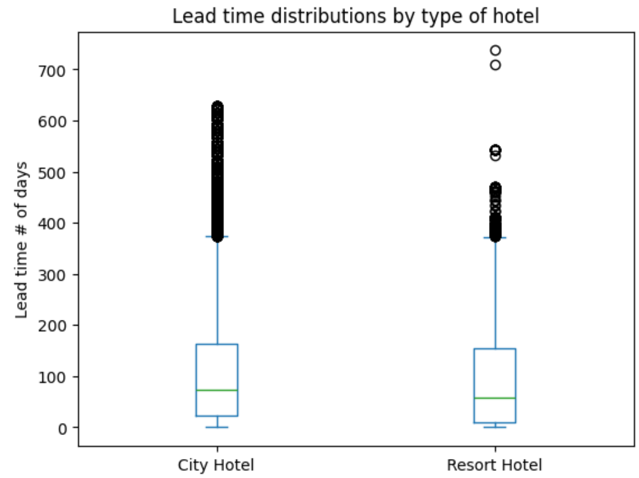
Before selecting predictor variables and creating a model, it is important to be familiar with the make-up of the data set. The data contains information on individual reservations at two hotels, a city hotel and a resort hotel. The city hotel makes up two-thirds of the data points and the other third is the resort hotel. Of all the data points in the set, 37% of reservations were canceled. In addition, of all reservations, nearly 82% of them were booked through a travel agent or tour operator. The temporal range of the data set spans July 2015 to August 2017, spanning a full 26 months. This will account for variety in booking and cancellation rates due to seasonal changes. Further, there were no missing values that needed to be handled, although a few categorical variables do include an “undefined” option. However these are quite a small percentage of all data points. For example, in the attribute distribution_channel, 5 of the points are labeled as undefined, representing an insignificant percentage of the overall data set.

Based on the literature, it was important to evaluate the relationship between cancellations and lead time. Lead time is the number of days that elapsed between the booking and the arrival date. The lead time is left-skewed with a large percentage of reservations made with a short lead time. For instance, 14% of all reservations were made with 5 days or less lead time. On the other hand, only 6% of reservations are made with more than 300 days lead time. This is further illustrated in the histograms below.



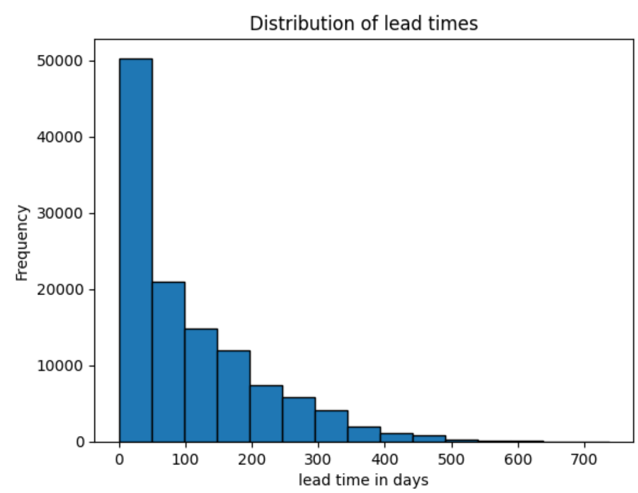


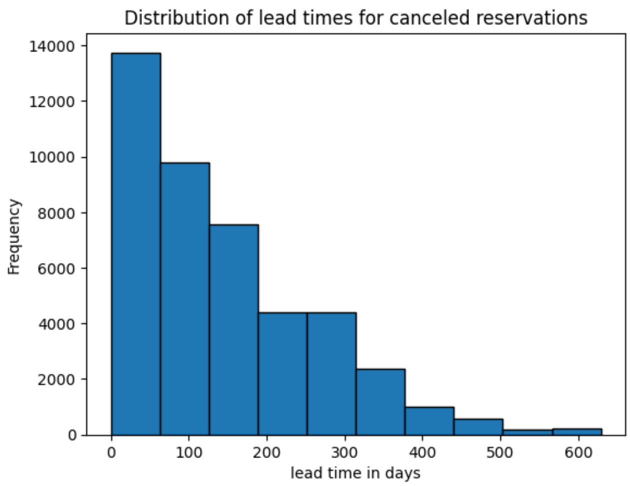
The lead time distributions between the two hotel types are generally shorter for the city hotel, although they are still fairly close. This is further seen in the boxplot and quantile distributions below.



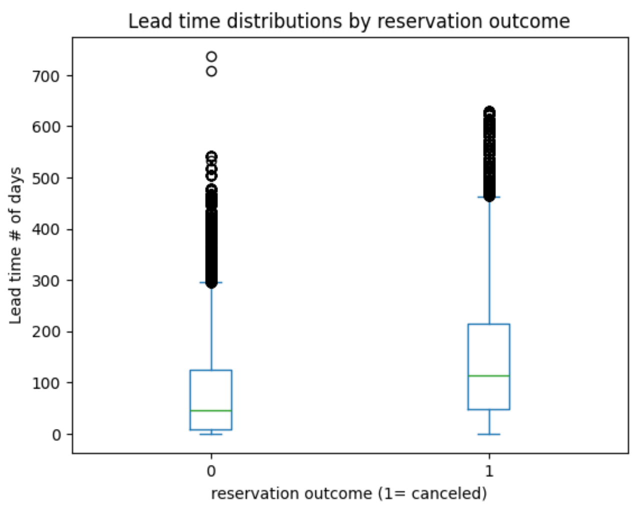
Quartile for lead_time	City Hotel	Resort Hotel
1st (0.25)	23	10
2nd (0.5)	74	57
3rd (0.75)	163	155

The distribution of lead times for reservations that were canceled is fairly different, as can be easily seen by comparing their histograms.





This can also be seen by comparing the distribution of lead times for reservations that were canceled versus those that were kept. This comparison shows reservations that result in cancellations skew more towards longer lead times compared to the reservations that followed through.



There is a positive correlation between lead time and cancellations at 0.293. Removing the shorter lead times lowers this correlation. For example, the correlation between cancellations and lead times greater than 5 days is 0.23.

The distribution channel variable indicates whether a reservation came through a travel agent, corporation or directly through the individual. In its raw format, this is a categorical variable. In order to determine its association with the is_canceled variable it was label encoded and then applied the chi-square test. This test indicates a strong association between cancellations and the distribution channel variable.

chi-squared statistic:	3745.79
p-value:	0
degrees of freedom:	4
expected frequencies:	[[4.20373048e+03 9.22025354e+03 1.21509657e+02 6.16173584e+04 3.14791859e+00] [2.47326952e+03 5.42474646e+03 7.14903426e+01 3.62526416e+04 1.85208141e+00]]

This data exploration sets us up to move into the next stages of refining our investigation and selecting predictors to begin modeling efforts. The attributes analyzed thus far are a good starting point of interesting factors that could be used to predict cancellations.

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